Memory for All

SAGE: Spatial Associative Geometric Embeddings

A Gradient-Free Geometric Memory Architecture with Hippocampal-Inspired Consolidation

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Abstract

Every time an AI system needs to learn something new, it must retrain from scratch — consuming months of compute and significant resources, while still forgetting what it knew before. The cause is architectural: knowledge is stored inside weight matrices that cannot be updated without destroying existing information. This paper proposes storing knowledge as geometric positions in a 3D cube instead — positions that can be updated instantly, never overwrite each other, and require no gradient computation to modify.

This paper introduces SAGE — Spatial Associative Geometric Embeddings — a memory architecture that stores knowledge as coordinate positions in a 3D geometric cube rather than in weights. Retrieval uses cosine similarity. Learning uses local activation-weighted prototype updates (Hebbian in locality). No backpropagation is required at any stage.

The result is a system with properties that parametric memory cannot provide: SAGE learns continuously without retraining, forgets 92% less than vanilla neural networks and 91% less than EWC when new information is added, maintains perfect retention across 200 sequential concept updates, and achieves 0.000% output activation sparsity — confirming geometric output localisation. A density stress test demonstrates 100% top-1 retrieval accuracy from 5% to 95% cube utilisation. A scaling ablation confirms that horizontal MultiCube scaling maintains higher response quality than vertical scaling at equivalent load.

Six contributions are made, as described in Section 1: gradient-free 3D coordinate storage; self-organising anti-collision; SAGEDivided working memory with fixed spatial partition; MultiCube horizontal scaling; a hippocampal-inspired consolidation pathway; and SAGESequenceCube — explicit geometric transition memory achieving 100% rollout accuracy — all without backpropagation.

Together these components implement, to the best of the author's knowledge for the first time in a gradient-free system, the two-stage hippocampal memory model that neuroscience has confirmed in the human brain. A companion paper demonstrates SAGE powering an autonomous agent that continued operating — zero defaults across 72 steps — even when its language model was disabled.

The geometry computes. The weights are not needed.

Code: https://github.com/Ivelin2022/sage

1. Introduction

Modern AI systems store all knowledge inside learned weight matrices. This creates three fundamental problems. First, updating knowledge requires retraining — a process that costs substantial compute and time. Second, learning new information overwrites old information through catastrophic forgetting (Kirkpatrick et al., 2017). Third, systems have no persistence mechanism when the computational substrate becomes unavailable.

Memory-augmented architectures such as Neural Turing Machines (Graves et al., 2014), Differentiable Neural Computers (Graves et al., 2016), and retrieval-augmented generation (Lewis et al., 2020) address these problems by separating compute from memory. More recent systems — including kNN-LM (Khandelwal et al., 2020), LongMem (Wang et al., 2023), and MemoryLLM (Wang et al., 2024) — demonstrate that external memory can be updated without full model retraining. However, these systems either require backpropagation through the memory access mechanism, rely on a learned controller, use static external databases that do not update during inference, or require the language model to remain online to process retrieved context. None provides the combination of fixed-address geometric storage, gradient-free writes, a hardcoded spatial partition encoding relational roles, and a structural guarantee against catastrophic forgetting.

This work proposes a different approach: store knowledge as positions in a three-dimensional geometric space, retrieve by cosine similarity, and learn through local activation-weighted prototype updates with no gradient computation. The geometry is not a visualisation aid — it is the computational substrate. I refer to this architecture as SAGE: Spatial Associative Geometric Embeddings.

SAGE makes the following contributions:

(1) A gradient-free 3D geometric memory where knowledge is stored as coordinate positions and retrieved by cosine similarity over full embedding vectors, with no learned parameters in the retrieval mechanism.

(2) Self-organising anti-collision via cosine-directed label assignment — new knowledge gravitates to the most compatible existing grid point through cosine similarity in the full embedding space, preventing interference without explicit collision management, learned gating, or vigilance parameters.

(3) SAGEDivided — a short-term working memory with a fixed axis-aligned spatial partition encoding relational roles (subject: $x < 0$, object: $x \geq 0$) geometrically, without learned transformations.

(4) MultiCube — a long-term memory that scales to arbitrary capacity by adding fixed-size specialist cubes, maintaining per-cube retrieval quality independent of total system size.

(5) A consolidation pathway modelled on hippocampal-neocortical memory transfer that continuously moves associations from SAGEDivided to MultiCube without backpropagation, firing every inference step.

(6) SAGESequenceCube — a specialist cube for explicit episodic transition memory, separating the retrieval index from the association pointer. Achieves 100% top-1 retrieval and 100% multi-step rollout accuracy with zero catastrophic forgetting after 300 noise transitions.

Of these six contributions, (1) and (3) represent fundamental architectural novelties with no direct precedent in the literature: no prior system stores knowledge as literal 3D coordinate positions rather than as weight values, and no prior gradient-free architecture uses a fixed axis-aligned spatial partition to encode relational roles geometrically. Contributions (2), (4), (5), and (6) represent novel compositions: the anti-collision mechanism applies nearest-prototype assignment (Carpenter and Grossberg, 1987) without vigilance parameters or weight updates; horizontal cube scaling applies the fixed-capacity modular principle (He, 2024) to memory rather than compute; the consolidation pathway is, to the best of the author's knowledge, the first gradient-free computational implementation of the complementary learning systems theory (McClelland et al., 1995); and SAGESequenceCube resolves the retrieval-association conflict in geometric sequence memory by separating embedding reinforcement from explicit dictionary lookup — achieving perfect rollout accuracy where prior sequence encoding approaches failed. The local prototype update rule and softmax retrieval are established mechanisms used as computational infrastructure, not claimed as novel.

This paper demonstrates structural anti-forgetting (92% less forgetting than neural network baseline, 91% less than EWC), continuous learning over 200 sequential concepts with perfect retention on a 16^3 cube, geometric localisation yielding 0.000% output activation sparsity on a 32^3 cube with GloVe 6B embeddings, 100% top-1 retrieval accuracy from 5% to 95% cube utilisation, and superior retrieval quality under horizontal versus vertical scaling at high load.

2. Related Work

2.1 Associative Memory

Sparse Distributed Memory (SDM; Kanerva, 1988) is the closest conceptual ancestor of SAGE. SDM stores binary patterns at fixed random address points in high-dimensional Hamming space, retrieving by threshold-based proximity. SAGE differs fundamentally in three ways: knowledge is stored in 3D Euclidean space rather than ~1000-dimensional binary space; knowledge is the position itself rather than a counter value stored at an address; and retrieval uses continuous cosine similarity rather than Hamming distance thresholding. Bricken and Pehlevan (2021) proved that softmax attention approximates SDM, providing a theoretical bridge between SAGE's softmax retrieval and associative memory theory.

Modern Hopfield Networks (Ramsauer et al., 2021) achieve exponential storage capacity through high-dimensional energy landscapes. They require energy-based convergence and operate on synchronous pattern updates. SAGE uses no energy function, no synchrony, and no weight matrices. Adaptive Resonance Theory (Carpenter and Grossberg, 1987) introduced similarity-directed category assignment — new inputs activate the most similar existing prototype. ART requires a vigilance parameter, dynamic category creation, and iterative weight updates on resonance; SAGE uses none of these.

2.2 Memory Scaling

Memory Layers at Scale (Berges et al., 2024) scales key-value memory to billions of entries within a single enlarged table. UltraMem (Huang et al., 2025) reaches 120 billion parameters through a similar vertical enlargement strategy. Both approaches increase capacity by growing a single store. SAGE scales horizontally — adding fixed-size specialist cubes — maintaining constant per-cube density and retrieval quality as a function of domain, not total size. PEER (He, 2024) scales compute modules horizontally using product-key retrieval, but its modules are functional transformers, not persistent memory stores.

2.3 Geometric and Spatial Neural Architectures

Achterberg et al. (2023) embedded recurrent neural networks spatially in 3D, demonstrating that spatial embedding improves generalisation. Their networks retain learned weight matrices; positions regularise but do not replace computation. Erb et al. (2025) trains neural networks by optimising neuron positions, deriving weights from inverse 3D distance. SAGE does not derive weights from positions — it stores knowledge as positions directly, eliminating weights from the storage-retrieval loop entirely. Self-Organising Maps (Kohonen, 1982) find a Best Matching Unit on a 2D grid through iterative neighbourhood weight updates; SAGE uses a fixed 3D grid with no weight adaptation during retrieval.

2.4 Relational Geometric Encoding

TransE (Bordes et al., 2013) encodes knowledge graph relations as translation vectors in embedding space. SAGE extends this principle to a physical 3D partition: the x-axis divides the cube into subject ($x < 0$) and object ($x \geq 0$) halves, encoding relational roles geometrically without learned transformations. BoxE (Abboud et al., 2020) uses axis-aligned hyper-rectangles for knowledge graph head/tail regions, but these boxes are fully learned through training. SAGE's partition is fixed a priori — hardcoded, not optimised.

2.5 Hippocampal Memory Models

The complementary learning systems (CLS) theory (McClelland et al., 1995) proposes that the hippocampus rapidly encodes episodic memories which are subsequently consolidated into neocortical long-term storage. Daume et al. (2024) demonstrated empirically that hippocampal working memory activity predicts long-term memory formation, confirming a shared cellular substrate. Panichello and Buschman (2021) showed that multiple items in working memory are maintained in distinct neural subspaces to minimise interference. SAGE implements this two-stage model computationally: SAGEDivided as the hippocampal rapid-encoding working memory with geometrically separated subspaces, MultiCube as the neocortical long-term specialist stores, and the consolidation pathway as the transfer mechanism — all without backpropagation. To the best of the author's knowledge, no prior gradient-free architecture is known to implement this two-stage model.

3. Architecture

3.1 Base SAGE: 3D Geometric Storage

A SpatialCube consists of N^3 grid points with coordinates $(x, y, z) \in [-1, +1]^3$. Each point p_i stores a full d-dimensional embedding vector $e_i \in \mathbb{R}^d$. Positions are assigned at initialisation on a regular grid and never change — they are the permanent spatial address of each memory slot.

Retrieval.

Given a query vector $q \in \mathbb{R}^d$, all stored embeddings and query vectors are L2-normalised before similarity computation. Softmax attention with temperature $\tau = 0.1$ concentrates activation on the top-k most similar points:

```
scores = softmax( (E · q) / τ )   [E, q are L2-normalised; dot product equals cosine similarity]
response = Σ scores_i · e_i   (over top-k)
```

where $E \in \mathbb{R}^{(N^3 \times d)}$ is the normalised embedding matrix. No learned parameters appear in retrieval. The geometry — specifically the cosine similarity structure of the stored embeddings — computes the answer. Retrieval scores all N^3 embeddings per query; the softmax-weighted response concentrates on the geometrically nearest neighbourhood (output localisation), but compute is dense.

Learning.

Learning uses an activation-weighted local prototype update (Hebbian in locality): activated points are pulled toward the target embedding, weighted by their softmax activation score:

```
e_i ← e_i + α · score_i · (t - e_i)
e_i ← e_i / ‖e_i‖   (renormalise)
```

where $t \in \mathbb{R}^d$ is the target embedding and α is the learning rate. No backpropagation is required. Gradients are accumulated across a batch and applied as a single averaged update to prevent oscillation.

Spatial cohesion (Force 1 — base architecture).

Every five learning steps, a spatial cohesion force pulls 3D-nearby points together if they are already semantically similar (cosine similarity > 0.3). The force magnitude is $\beta \cdot$ cosine_similarity. β is scaled by cube size: β = G_base $\cdot$ (cube_size / 32)^3 where G_base = 0.003. This cube-size scaling is required because neighbourhood count grows as n^3 — without it, smaller cubes would experience proportionally stronger cohesion.

3.2 Extended Learning Forces (V2 Architecture)

Four additional forces are introduced in the V2 architecture, which is used in all multi-step experiments. These forces operate on the embedding vectors during learning and do not affect the retrieval mechanism.

Momentum (Force 2).

A velocity term accumulates across learning steps, preventing oscillation between conflicting targets:

```
v_i ← γ · v_i + (1 − γ) · score_i · (t − e_i)
e_i ← e_i + α · v_i
```

where $\gamma = 0.9$ is the momentum coefficient. This is the same principle as SGD with momentum — directional consistency is rewarded, oscillation is damped.

Contrastive repulsion (Force 3).

A random sample of non-target points is pushed away from the target direction after each batch:

```
e_j ← e_j − α · neg_weight · (t − e_j)   for j ∉ top-k
```

where neg_weight = 0.3. Without this force, representations collapse — all points drift toward the same targets. Contrastive repulsion is the mechanism behind V2's 20% cluster cohesion improvement over V1 (Table 4).

Direction training (Force 4).

In addition to learning the target embedding, the system learns the direction vector $(t - q)$ between query and target. This directly enables analogy arithmetic: the relationship between two concepts is encoded as a geometric direction, not just two point locations. The formal update for each training pair (query q, target t):

```
direction vector:  d = normalize(t − q)
for each point i with activation score s_i > threshold:
    e_i ← normalize(e_i + α_dir · s_i · d)
```

This extends the TransE principle (Bordes et al., 2013) to the SAGE embedding update. Direction training successfully encodes relational structure in the cube geometry during learning; however, its benefit is sensitive to training epoch count and relation type. On the 12-item curated evaluation set (Table 3), direction training improves accuracy from 16.7% to 58.3%. On the full Google analogy benchmark (6,762 questions, 10K vocabulary restriction), performance requires the caveat described in Section 4.5.

Adaptive temperature (Force 5).

The softmax temperature τ decreases from $\tau\_start = 0.3$ to $\tau\_end = 0.05$ over the first 500 learning steps:

```
τ(t) = τ_end + (τ_start − τ_end) · max(0, 1 − t/500)
```

High temperature early in training (broad exploration) gradually sharpens to low temperature (precise retrieval). This prevents premature convergence to local regions of the cube before the global semantic structure has formed.

3.3 Lennard-Jones Gravity (V3/V4 Architecture)

The V3 architecture replaces the base spatial cohesion term with a full Lennard-Jones-inspired force field (Lennard-Jones, 1924) acting in 3D position space. The V3/V4 architecture relaxes the fixed-position constraint during training only: positions are initialised on a regular lattice and updated by the spatial force field during training; after training completes, positions are frozen for inference. This is distinct from the base architecture (V1/V2) where positions are fixed throughout. The actual field uses cosine-gated Gaussian falloff (not the standard r^{-12}/r^{-6} form). Two sub-forces operate simultaneously:

Attraction.

Points with positive cosine similarity (similar concepts) attract each other in 3D space:

```
F_attract(i,j) = G · cosine(e_i, e_j) · exp(−d_ij / σ) · pulse(t) ·
cf(p_i) · cf(p_j)
```

where d_ij is the 3D Euclidean distance, $\sigma = 0.5$ is the Gaussian falloff radius, $G = 0.003$ is the gravity constant scaled by cube size, and $cf(p) = 1 - \max(|x|, |y|, |z|)^2$ is the corner dampening factor.

Repulsion.

Points with negative cosine similarity (dissimilar concepts) repel each other:

```
F_repel(i,j) = G · repel_factor · |cosine(e_i, e_j)| · exp(−d_ij / σ) ·
pulse(t) · cf(p_i) · cf(p_j)
```

where repel_factor = 0.6.

Oscillatory pulse.

The gravity force breathes sinusoidally over training steps:

```
pulse(t) = 1 + A · sin(ω · t)
```

where $A = 0.4$ and $\omega = 0.05$ rad/step. This periodic relaxation prevents semantic clusters from freezing in incorrect configurations early in training — concepts can escape wrong neighbourhoods during the low-gravity phase of each cycle.

Corner dampening.

Points near the cube corners receive weakened gravity: $cf(p) = 1 - \max(|x|, |y|, |z|)^2$. Without this, the bounded cube geometry creates artificial attraction toward corners where fewer neighbours exist. Corner dampening ensures the force field is approximately uniform across the interior of the cube.

3.4 Self-Organising Anti-Collision

A central challenge in fixed-capacity memory systems is collision: two semantically unrelated concepts assigned to the same storage slot. SAGE avoids collision through cosine-directed label assignment. When new knowledge k is stored, the system queries the cube with k's embedding and assigns k to the point with the highest cosine similarity:

```
p* = argmax_i  cosine(e_i, embed(k))
```

Semantically distinct concepts produce near-orthogonal embedding vectors, which score highest at different grid points. Collision probability between two concepts is therefore proportional to their semantic similarity — which is exactly when sharing a neighbourhood is correct behaviour. Semantically similar concepts occupy the same region; dissimilar concepts self-segregate. This mechanism requires no vigilance parameter, no dynamic category creation, no reset mechanism, and no learned routing function.

3.5 SAGEDivided: Short-Term Working Memory

SAGEDivided implements a short-term working memory with a structural innovation: the cube is permanently partitioned along the x-axis. Points with $x < 0$ encode subjects (what is being perceived); points with $x \geq 0$ encode objects (what action was taken or what response was produced). This partition is hardcoded at initialisation and never changes.

The relational structure between a perception and its associated response is therefore preserved as geometry: the direction vector from a stored subject to its corresponding object encodes the semantic relationship. This extends the TransE analogy arithmetic principle (Bordes et al., 2013) to a physically partitioned space: relations are geometric directions crossing a fixed boundary, not learned translations.

The partition provides a structural guarantee against subject-object interference — no subject concept can ever occupy the object half, regardless of its embedding content. This is directly analogous to the neuroscientific finding that working memory items are maintained in distinct neural subspaces to minimise interference (Panichello and Buschman, 2021).

3.6 MultiCube: Long-Term Specialist Memory

MultiCube comprises N specialist SpatialCubes, each responsible for a distinct knowledge domain. In a typical configuration, each cube covers one domain — for example: factual knowledge, relational context, temporal history, and task goals. The number of cubes and their domains are defined at deployment time and can be extended without modifying existing cubes.

The key architectural principle is horizontal scaling: when knowledge capacity must increase, a new specialist cube is added rather than enlarging existing cubes. Each cube maintains constant size (16^3 = 4,096 points in a typical configuration), constant density, and therefore constant retrieval quality. This contrasts with vertical scaling approaches (Memory Layers at Scale; UltraMem) where a single store grows without bound, degrading the signal-to-noise ratio of retrieval at high density. An operational constraint governs each specialist cube: retrieval quality degrades as occupancy exceeds approximately 50% of available grid points. Each MultiCube specialist is maintained below this threshold; a new specialist cube is allocated when this ceiling is approached.

Retrieval quality in MultiCube is formally domain-bounded: queries to one specialist cube compete only against memories within that cube, not against all memories in the system. Softmax concentration remains sharp regardless of total system knowledge.

3.7 Consolidation Pathway

The consolidation pathway transfers associations from SAGEDivided (working memory) to MultiCube (long-term memory) after every inference step. This continuous transfer is modelled on the hippocampal-neocortical consolidation process (McClelland et al., 1995; Daume et al., 2024).

After each query-response pair is processed through SAGEDivided, the subject embedding is stored in the appropriate specialist cube of MultiCube using the same cosine-directed label assignment mechanism. The working memory is not cleared — it accumulates context across the current interaction episode. Long-term memory grows continuously with every step. No batch collection, no replay buffer, no sleep cycle is required.

3.8 SAGESequenceCube: Explicit Transition Memory

The base SAGE architecture is order-agnostic — the same query vector produces the same retrieval regardless of step position. SAGESequenceCube extends SAGE with explicit transition memory: given any state A, the system can retrieve the geometrically stored next state B, and chain these transitions to roll out a full sequence A→B→C→D without an LLM.

Earlier approaches to sequence awareness (sinusoidal positional encoding, delta encoding) modified the query vector at retrieval time. This conflated two responsibilities: the retrieval index (finding the right point) and the association (knowing what comes next). SAGESequenceCube separates them cleanly:

Retrieval index — embeddings are trained to stay close to their input vectors. Cosine search remains reliable: querying with state A finds the correct subject grid point.

Association — an explicit dictionary stores the s_idx→o_idx pointer. This is collision-free and independent of embedding geometry.

The subject/object spatial partition (Section 3.5) is preserved: subject-half points ($x < 0$) store current states; object-half points ($x \geq 0$) store next states. Retrieval proceeds as: find nearest subject-half point to query (retrieval index) → look up dictionary → return associated object-half point (association). The geometry handles search; the dictionary handles the relational link.

Experiment	Result	Interpretation
Exp 1: Retrieval accuracy	Top-1: 100% (12/12)	Correct next-state found every time
Exp 2: Multi-step rollout	100% (12/12 steps)	A→B→C→D chains without error
Exp 3: Spatial clustering	0.987x — Weak	Collision-free but not spatially clustered
Exp 4: Anti-forgetting	0.0% forgetting	300 noise transitions leave existing memories intact

Table 7: SAGESequenceCube v5 results (32^3 cube, 64d embeddings, 80 epochs, α=0.2). Perfect retrieval and rollout accuracy. Spatial clustering is weak — collision avoidance is structural but spatial organisation of sequences is not. Anti-forgetting is perfect.

Experiment 3 reveals an honest limitation: same-sequence states occupy collision-free but spatially unorganised positions (0.987x separation ratio). The partition prevents interference but does not impose spatial locality on sequences. This distinguishes SAGESequenceCube from the broader SAGE spatial organisation claim — here, geometry provides collision avoidance, not clustering.

4. Experiments

4.1 Setup

Baseline comparison (Section 4.2) and architecture progression (Section 4.7) use a $32^3 = 32{,}768$ point SpatialCube with 64-dimensional embeddings on CUDA. Anti-forgetting and continuous learning experiments (Sections 4.3–4.4) use a $16^3 = 4{,}096$ point cube with 64-dimensional embeddings. GloVe analogy evaluation (Section 4.5) uses a 32^3 cube with 50-dimensional embeddings. Load sweep and scaling ablation experiments (Sections 4.10–4.11) use a 32^3 cube with 64-dimensional embeddings. Hardware: NVIDIA RTX 4090 Laptop GPU, CUDA backend, PyTorch.

For word embedding experiments, this work uses GloVe 6B 50-dimensional vectors (Pennington et al., 2014) with 10,000 vocabulary items. V1 analogy experiments train on 2,060 pairs for 200 epochs. V2 direction training experiments train on 40,036 pairs — the full GloVe 10,000 vocabulary pair set expanded across 200 epochs. The difference in training volume partly explains the V1→V2 accuracy improvement on the curated analogy set.

4.2 Baseline Comparison: SAGE vs MLPs

Table 1 reports SAGE against MLP-Small (99,904 parameters) and MLP-Large (2,237,504 parameters) on an embedding mapping task with 420 pairs in 64 dimensions on CUDA.

Metric	MLP-Small	MLP-Large	SAGE (32^3)
Parameters	99,904	2,237,504	2,195,456
Training time (s)	4.71	4.50	9.72
Start loss	0.881	0.848	0.379
Final loss	0.216	0.191	0.309
GPU memory (MB)	19.2	N/A	41.2
Single query QPS	3,671	2,264	3,333
% params active	100%	100%	0.000%
Continuous learning	No	No	Yes
Anti-forgetting	No	No	Yes

Table 1: SAGE vs MLP baselines. SAGE achieves 0.000% output activation sparsity (geometric output localisation), continuous learning, and structural anti-forgetting unavailable in either MLP baseline.

SAGE starts with a substantially lower initial loss (0.379 vs 0.848) reflecting its geometric prior — the cube begins with meaningful structure before any learning. MLP-Large achieves lower final loss (0.191 vs 0.309), consistent with a parametric model that can fit the training distribution. SAGE trades fitting accuracy for three capabilities unavailable in MLPs: geometric output localisation, continuous learning without retraining, and structural anti-forgetting.

4.3 Catastrophic Forgetting

SAGE is compared against a vanilla neural network and Elastic Weight Consolidation (EWC; Kirkpatrick et al., 2017) on a two-task sequential learning protocol. All systems first learn Task A (420 pairs), then learn Task B (420 different pairs). Retention of Task A is measured after Task B learning. EWC uses Fisher information matrix regularisation (best λ=10000, reported from λ sweep: 100/1000/10000). Results are averaged over 3 runs ± std. Forgetting is measured as the reduction in Task A cosine similarity between post-Task-A and post-Task-B evaluation; the comparison is of relative retention loss.

System	Forgetting (mean ± std)	Task A before	Task A after	Reduction vs Vanilla
Vanilla MLP	0.669 ± 0.014	0.998	0.328	—
MLP + EWC (λ=10000)	0.553 ± 0.012	0.997	0.445	17.4%
SAGE V2	0.049 ± 0.005	0.051	0.002	92.7%

Table 2: Three-way catastrophic forgetting comparison (420 pairs/task, 64d, 3 runs ± std). SAGE achieves 92.7% less forgetting than vanilla MLP and 91.2% less than EWC (best λ=10000). EWC reduces but cannot eliminate forgetting because shared weight matrices remain. SAGE's structural geometric separation provides a categorical rather than continuous improvement. Note: SAGE's absolute cosine similarity values are lower than MLP because Hebbian updates partially move embeddings — the comparison is of relative retention loss, not absolute level.

SAGE's forgetting (0.049) is 92.7% lower than the neural network's forgetting (0.669) and 91.2% lower than EWC's best result (0.553). The retention curve shows why EWC cannot match SAGE on this task: EWC slows forgetting but cannot stop it — both MLP and EWC+MLP curves continue declining throughout 200 epochs, just at different rates. SAGE's structural geometric separation is a guarantee, not a regularisation penalty.

4.4 Continuous Learning

200 distinct concepts are streamed to SAGE one at a time, tracking retention of the first 10 concepts throughout the stream. Retention is measured as the cosine similarity between the stored embedding and the original concept vector.

SAGE maintains perfect retention (1.0000) for all 200 steps across all 10 tracked concepts. The geometric self-organisation mechanism ensures each new concept occupies its own neighbourhood without displacing earlier memories. This result demonstrates that SAGE's anti-forgetting is not a function of low memory density — it holds across a 200-step stream representing 4.9% cube utilisation.

4.5 Real-World Embeddings: GloVe Analogy Test

SAGE is trained on GloVe 6B 50d embeddings and evaluated on word analogy tasks. Table 3 reports accuracy on 12 curated analogy pairs. Note: Table 3 reports V1 at 16.7% (2/12) on this curated 12-item set; the 33.3% figure sometimes cited elsewhere is from a separate V1 run with a different 50d GloVe training configuration. The curated 12-item set is the canonical evaluation for version comparison throughout this paper.

Analogy	GloVe	SAGE V1	SAGE V2
king - man + woman = queen	✓	✓	✗
queen - woman + man = king	✓	✗	✓
paris - france + italy = rome	✓	✓	✗
berlin - germany + france = paris	✓	✗	✓
london - england + france = paris	✓	✗	✓
rome - italy + germany = berlin	✓	✗	✓
walking - walk + swim = swimming	✓	✗	✗
walked - walk + run = ran	✓	✗	✗
bigger - big + small = smaller	✗	✗	✓
faster - fast + slow = slower	✓	✗	✓
actor - man + woman = actress	✓	✗	✓

Analogy	GloVe	SAGE V1	SAGE V2
uncle - man + woman = aunt	✗	✗	✗

Table 3: Curated analogy accuracy (12 pairs). GloVe: 83.3% (10/12). SAGE V1: 16.7% (2/12). SAGE V2 (direction training, arithmetic query): 58.3% (7/12). Direction training closes 62% of the original gap on this curated set. The remaining gap reflects training data volume — GloVe encodes structure from billions of co-occurrences; SAGE V2 trains on 40,036 pairs.

GloVe achieves 83.3% accuracy (10/12) vs SAGE V1 at 16.7% (2/12). SAGE V1 correctly resolves only analogies where the answer is a direct geometric neighbour of the query. With V2 direction training (Force 4) and arithmetic querying (a - b + c), accuracy rises to 58.3% (7/12) — closing 62% of the original gap. The remaining gap reflects training data volume: GloVe encodes semantic structure from billions of co-occurrences; SAGE V2 trains on 40,036 pairs. Both fail on morphologically complex pairs (bigger/smaller, uncle/aunt).

Full Google analogy benchmark (6,762 questions, 10K vocabulary restriction): GloVe achieves 3.4% top-1 (10.5% top-5); SAGE V1 achieves 2.3% top-1 (6.3% top-5); SAGE V2 achieves 0.7% top-1 (2.4% top-5). These low absolute numbers reflect the vocabulary restriction — many correct answers are outside the 10K candidate set, penalising all systems equally. GloVe evaluated on its full 400K vocabulary achieves ~75%. On the restricted vocabulary, SAGE V1 reaches 67% of GloVe's performance (2.3% vs 3.4%). SAGE V2 performs lower than V1 on the full benchmark at 200 epochs — direction training improves relational arithmetic on the curated in-distribution set but reduces general retrieval quality at this training duration, indicating sensitivity to training epoch count.

4.6 SAGEDivided + MultiCube: Architectural Ablation

An ablation study comparing MultiCube alone (v1) against SAGEDivided + MultiCube + Consolidation (v2) is reported in the companion paper (Likov, 2026b), which applies SAGE to autonomous drone navigation as a use case. In that study, 72 agent steps across 12 scenarios with the LLM disabled after step 8 show that adding SAGEDivided working memory improves SAGE-only meaningful recall by 60% (5/12 to 8/12) at a cost of 2x response time and 25% memory footprint increase. The consolidation pathway ran on all 12 steps without failure. Zero default decisions were recorded in either version. These results are presented in full in the companion paper and not reproduced here to avoid redundancy.

4.7 Architecture Progression: V1 → V2 → V3 → V4

Table 4 reports the four-way comparison across SAGE versions on synthetic concept vectors (10^3 cube, 64-dimensional embeddings). All versions use identical training data, seeds, and evaluation protocols. V1 is the base architecture. V2 adds momentum, contrastive repulsion, direction training, and adaptive temperature. V3 adds the similarity-gated spatial force field with oscillatory pulse and corner dampening over V1 learning. V4 combines V2 learning with V3 gravity.

Metric	V1	V2	V3	V4
Cluster cohesion (higher = better)	1.47x	1.77x	1.46x	1.75x
Anti-forgetting Δloss (lower = better)	-0.014	-0.000	-0.014	-0.000
Analogy accuracy (synthetic)	0%	0%	0%	0%
Sparsity	0.000%	0.000%	0.000%	0.000%

Table 4: Four-way architecture comparison on synthetic data (10^3 cube). V2 and V4 achieve perfect anti-forgetting (-0.000) and 20% better cluster cohesion vs V1/V3. Analogy accuracy is 0% on synthetic data — this requires real semantic vectors (see Table 3).

The key findings are: (1) V2's contrastive repulsion drives cluster cohesion improvement — V3 gravity alone adds negligible cohesion over V1 (1.46x vs 1.47x); (2) V2 and V4's continuous_learn() no-momentum mode achieves perfect anti-forgetting (-0.000) while V1 and V3 show measurable drift (-0.014); (3) V4 preserves V2's gains while adding spatial self-organisation through gravity; (4) sparsity is 0.000% across all versions confirming it is a structural geometric property independent of the learning rule.

On real GloVe 6B embeddings (32^3 cube, 50-dimensional), V1 achieves 33.3% analogy accuracy and V4 matches at 33.3% — a draw. The gravity contribution to analogy arithmetic does not manifest at this training scale. With V2 direction training (Force 4), accuracy rises to 58.3% on the curated 12-item set (Table 3), demonstrating that the limiting factor for the curated pairs was the learning rule, not the cube architecture or scale.

4.8 Sequence Awareness: Encoding Mechanisms

SAGE's base retrieval is order-agnostic — the same query vector produces the same result regardless of step position in a sequence. Three sequence encoding mechanisms are implemented and evaluated: positional encoding (sinusoidal position vector added to the query), delta encoding (direction-of-change vector from previous to current observation added to the query), and combined encoding (both together at reduced weight). Each mechanism modifies the query vector before retrieval without changing the stored memory:

```
positional:  q' = normalize(q + 0.3 · sin_pos(t))
delta:       q' = normalize(q + 0.3 · (q_t - q_{t-1}))
combined:    q' = normalize(q + 0.2 · pos + 0.2 · delta)
```

Table 5 reports results across 12 agent navigation scenarios using a saved SAGE state with 12 seeded action memories on an NVIDIA RTX 4090 (CUDA, 768-dimensional embeddings, 16^3 = 4,096 point MultiCube).

Mode	Meaningful	Defaults	Quality	Confidence	Avg time
Baseline (no encoding)	12/12	0	100%	All low	2.11s
Positional	12/12	0	100%	All low	2.09s
Delta	12/12	0	100%	All low	2.10s
Combined	12/12	0	100%	All low	2.10s

Table 5: Sequence encoding comparison across 12 agent navigation scenarios. All four modes achieve 100% meaningful recall with 0 defaults. Sequence encoding neither improves nor degrades performance at this memory density.

All four modes achieve 100% meaningful recall with zero defaults. The inability to distinguish encoding effects reflects memory density rather than mechanism failure. With 12 seeded action pairs across 4,096 grid points (0.29% utilisation), retrieval confidence is uniformly low for all modes — the softmax distribution is too flat to reveal differences in query quality.

One qualitative observation is worth noting: on Step 8 ("Strong wind, agent drifting 2 metres per second east"), the positional mode returns a text decision ("Continue current mission") rather than a geometric point, while baseline, delta, and combined return point indices. This divergence — consistent across runs — suggests that positional encoding shifts the query toward a different semantic neighbourhood for time-indexed wind-related observations. It is reported here as an observed behavioural difference, not a quality claim.

4.9 Sequence Delta Encoding on GloVe Analogies

The delta encoding mechanism is additionally evaluated on the GloVe 50d analogy task. Three retrieval modes are compared on a freshly trained V2 cube (200 epochs, 40,036 pairs, 32^3 cube, CUDA): baseline query $q = c$, delta_ab query $q = \text{normalize}(c + w \cdot (b-a))$, and arithmetic query $q = \text{normalize}(a - b + c)$.

Mode	Accuracy	Notes
Baseline (q = c)	0/12 = 0%	Labels sit on query vectors — labelling artefact
Delta_ab best (w=0.5)	1/12 = 8.3%	Only walking+swim=swimming
Arithmetic (a - b + c)	7/12 = 58.3%	Direction training encodes relational structure

Table 6: Delta encoding on GloVe analogies. Arithmetic query significantly outperforms delta encoding. Direction structure is in the cube geometry but arithmetic is the correct retrieval mechanism.

Arithmetic querying achieves 58.3% — Force 4 (direction training) successfully encodes relational structure in the cube geometry during learning. Delta encoding at retrieval time achieves only 8.3% — the directional structure is present in the geometry but delta encoding at query time is a weaker access mechanism than direct arithmetic. This result clarifies the architecture: Force 4 (direction training) at learning time is essential; delta encoding at retrieval time is not the right complement. Standard analogy arithmetic is the correct retrieval approach when relational queries are known.

4.10 Load Sweep: Density Stress Test

A density stress test evaluates retrieval quality across cube utilisation from 5% to 95% — the range the reviewer identified as unanswered. A 32^3 cube (32,768 points) with 64d embeddings is filled to each level; concepts are random unit-norm vectors (near-orthogonal in 64D, making retrieval more discriminative than real semantic embeddings — a caveat for paper claims). For each level, top-1 accuracy, top-5 accuracy, collision rate, cosine similarity of retrieved vs stored, and query latency are measured. All N^3 embeddings are scored per query regardless of utilisation (output localisation, not compute sparsity).

Util %	Concepts	Top-1	Top-5	Collision	Cosine mean	Latency
5%	1,638	100%	100%	0.0%	0.986	0.075ms
10%	3,276	100%	100%	1.4%	0.948	0.093ms
20%	6,553	100%	100%	8.7%	0.935	0.080ms
40%	13,107	100%	100%	17.6%	0.929	0.091ms
60%	19,660	100%	100%	24.3%	0.929	0.100ms
80%	26,214	100%	100%	30.2%	0.930	0.082ms
95%	31,129	100%	100%	34.3%	0.932	0.117ms

Table 8: Load sweep across 5%→95% cube utilisation (32^3 cube, 64d synthetic, RTX 4090, 8 epochs). Top-1 and Top-5 retrieval accuracy remain at 100% across the full range. Response quality (cosine similarity) degrades gracefully and plateaus at 0.929 after 40% utilisation. Query latency is essentially constant. Collision rate rises linearly but does not affect retrieval accuracy — the Hebbian update mechanism resolves grid point sharing through embedding geometry. Experiment uses 64d near-orthogonal synthetic vectors; results with real semantic embeddings at high utilisation would show additional quality degradation.

Top-1 and Top-5 retrieval accuracy remain at 100% across all seven utilisation levels, including 95% where 34.3% of concepts share grid points. Response quality (cosine similarity) drops from 0.986 at 5% utilisation to 0.935 at 20%, then plateaus completely from 40% through 95%. Query latency is essentially constant (0.075–0.117ms) because the full N^3 cosine scan runs identically regardless of occupancy. Collision rate rises linearly but does not degrade retrieval accuracy — the Hebbian update resolves grid point sharing through the embedding geometry. The recommended 50% operational ceiling maintains cosine quality above 0.93 ± 0.015; above this, collision rate exceeds 17% while accuracy remains perfect.

4.11 Scaling Ablation: Vertical vs Horizontal

A scaling ablation compares two approaches to storing 13,107 concepts at equal total capacity (262,144 grid points each): vertical scaling uses a single 64^3 cube; horizontal scaling uses eight 32^3 cubes (MultiCube). Additionally, both approaches are tested at 40% per-cube utilisation (104,857 concepts total). Results are averaged over 3 runs.

Condition	Total pts	Util/cube	Top-1	Cosine	Latency
Vertical 32^3 (5% util)	32,768	5%	1.000	0.999	0.224ms
Vertical 32^3 (20% util)	32,768	20%	1.000	0.996	0.266ms
Vertical 32^3 (40% util)	32,768	40%	1.000	0.993	0.267ms
Horiz 2×32^3 (10% util/cube)	65,536	10%	1.000	0.985	0.261ms
Horiz 4×32^3 (10% util/cube)	131,072	10%	1.000	0.977	0.253ms
Vertical 64^3 (5% util)	262,144	5%	1.000	0.993	0.214ms
Horiz 8×32^3 (5% util/cube)	262,144	5%	1.000	0.981	0.233ms
Vertical 64^3 (40% util)	262,144	40%	1.000	0.931	0.207ms
Horiz 8×32^3 (40% util/cube)	262,144	40%	1.000	0.943	0.211ms

Table 9: Vertical vs horizontal scaling ablation (64d synthetic, 8 epochs, 3 runs, RTX 4090). Top-1 accuracy is 100% in all conditions. At low load (5%), both architectures perform equivalently. At high load (40% per cube), horizontal MultiCube maintains higher response cosine quality (+1.2%, 0.943 vs 0.931) than the vertical monolithic cube at equivalent total capacity. Latency is architecture-independent.

Top-1 retrieval accuracy is 100% in all ten conditions. At low load (5% per cube), both architectures perform equivalently — the MultiCube advantage is not needed when density is low. At high load (40% per cube), horizontal scaling maintains 1.2% higher response cosine quality (0.943 vs 0.931) than a monolithic cube at the same total capacity and utilisation. This confirms the architectural claim: domain-bounded softmax in specialist cubes prevents cross-domain interference that would otherwise flatten the softmax distribution. Latency is architecture-independent (0.207–0.267ms) because the GPU always scans N^3 points per cube in a single matrix multiply — adding cubes adds independent parallel queries, not sequential depth.

5. Discussion

5.1 Geometric Prior Advantage

SAGE begins with a substantially lower initial loss than MLPs (0.379 vs 0.848) because the cube is initialised with structure: random unit-norm embeddings at fixed grid positions already provide a meaningful cosine similarity landscape before any learning. This geometric prior — absent in MLPs that begin with random weights and no structure — accelerates early retrieval quality. It represents a form of inductive bias where the storage medium itself provides useful initial organisation.

5.2 The Gradient-Free Retrieval Guarantee

Standard neural memory systems (attention, MLP layers, memory networks) require learned weight matrices in the retrieval path. This couples the memory content to the retrieval mechanism: changing what is stored requires retraining the retrieval function. In SAGE, the retrieval function is fixed — cosine similarity against stored embeddings — and is independent of what is stored. New knowledge can be inserted by updating one embedding at one grid point without affecting any other retrieval operation. This decoupling is the source of both the continuous learning capability and the anti-forgetting guarantee.

5.3 How SAGE Avoids the Curse of Dimensionality

The curse of dimensionality (Bellman, 1961) states that in high-dimensional spaces, distance-based nearest-neighbour search degrades — all points become approximately equidistant, rendering proximity meaningless. This is a known weakness of Kanerva's SDM, which operates in ~1000-dimensional binary Hamming space and suffers dimensional collapse with correlated real-world data. SAGE addresses this through three architectural mechanisms.

First, the primary retrieval mechanism is cosine similarity over the full embedding dimension — not 3D Euclidean distance. The curse of dimensionality applies to distance-based search; cosine similarity measures angle rather than distance and remains discriminative in high-dimensional spaces. SAGE never uses 3D proximity as the primary retrieval signal. The 3D grid is a storage addressing system; the d-dimensional embedding space — 50D for GloVe experiments, 64D for synthetic experiments, and up to 768D in downstream applications — is where semantic computation occurs.

Second, MultiCube scales horizontally rather than vertically. The curse worsens as a single space grows larger — more dimensions, more volume, more empty space between points. SAGE sidesteps this by keeping each cube fixed at $16^3 = 4{,}096$ points regardless of total system knowledge. A new domain adds a new cube; no existing cube grows. Each cube remains in a density regime where retrieval quality is maintained. This is the architectural response to the curse: do not grow the space, add spaces.

Third, the SAGEDivided fixed partition imposes structure rather than discovering it geometrically. The subject/object split is hardcoded along the x-axis — it does not rely on the 3D geometry to learn or discover relational structure. By removing the need for 3D proximity to encode semantic roles, the partition avoids the geometric noise that the curse would introduce into any learned spatial organisation.

One component that is affected by the curse is the spatial cohesion term, which pulls 3D-nearby points together based on semantic similarity. In high-dimensional embedding space, points that are close in 3D are essentially random semantic neighbours — the 3D proximity and semantic proximity in embedding space are uncorrelated at initialisation. Spatial cohesion therefore adds noise rather than signal in the early training phase. This is why the cohesion force uses a high similarity threshold (cosine > 0.3) before activating — it only coheses points that have already converged semantically, not randomly nearby points.

In summary: SAGE avoids the curse for its primary retrieval mechanism (cosine similarity over the full embedding dimension d), responds architecturally to it for scaling (horizontal MultiCube addition), imposes structure rather than discovering it geometrically (fixed SAGEDivided partition), and acknowledges it as a residual limitation of the spatial cohesion term which is mitigated but not eliminated.

5.4 Limitations

MLP-Large achieves lower final loss than SAGE on the embedding mapping task (0.191 vs 0.309). This gap reflects the parametric advantage of a system that can fit the training distribution through gradient optimisation. SAGE is not designed to maximise fitting accuracy; it is designed to maximise updateability and retention. For tasks requiring high precision on a fixed dataset, parametric models remain superior.

SAGE V1 achieves 16.7% on the curated 12-item analogy set. With V2 direction training active, this rises to 58.3% on the curated set — closing 62% of the gap with GloVe (83.3%). On the full Google analogy benchmark (6,762 questions, 10K vocabulary restriction), SAGE V1 achieves 2.3% and SAGE V2 achieves 0.7% — both penalised by the vocabulary restriction that removes most correct answers from the candidate set. Direction training improves curated relational arithmetic but reduces general retrieval quality at 200 epochs, indicating sensitivity to training duration and relation type. Full compositional reasoning at GloVe parity requires either larger training sets, longer training with early stopping, or the SAGEDivided relational partition.

The density stress test (Section 4.10) uses 64d near-orthogonal synthetic vectors. Real semantic embeddings exhibit correlations that increase effective collision rates; results with GloVe embeddings at high utilisation would show additional quality degradation beyond the plateau observed with synthetic vectors.

All experiments use either synthetic concept vectors or GloVe embeddings. Integration with a perception encoder (vision, audio, sensor data) to produce 768D queries in real time is required for production deployment.

5.5 Comparison to Complementary Learning Systems Theory

The SAGEDivided + MultiCube + Consolidation architecture directly instantiates the biological two-stage memory model (McClelland et al., 1995). The hippocampus rapidly encodes episodic content in working memory (SAGEDivided); the neocortex gradually accumulates specialist knowledge through consolidation (MultiCube). Critically, SAGE implements this model without backpropagation in either stage — the first gradient-free computational instantiation of CLS theory the author is aware of.

6. Conclusion

This paper has presented SAGE, a gradient-free geometric memory architecture that stores knowledge as coordinate positions in a 3D unit cube, retrieved by cosine similarity without learned parameters. Two fundamental architectural novelties are introduced: gradient-free 3D coordinate storage with no precedent in prior memory systems, and SAGEDivided's fixed axis-aligned relational partition. Four novel compositions complete the system: cosine-directed self-organising anti-collision without vigilance parameters, MultiCube horizontal scaling applied to memory rather than compute, a consolidation pathway that is the first gradient-free implementation of the complementary learning systems model, and SAGESequenceCube — an explicit transition memory component achieving 100% retrieval and rollout accuracy through separated retrieval index and association mechanisms.

Empirical results demonstrate: structural anti-forgetting (92.7% less forgetting than a neural network baseline, 91.2% less than EWC); perfect retention across 200 continuous learning steps; 0.000% output activation sparsity confirming geometric output localisation; 100% top-1 retrieval accuracy from 5% to 95% cube utilisation in a density stress test; and horizontal MultiCube scaling maintains higher response quality (+1.2% cosine) than vertical scaling at high load (40%). On the curated 12-item analogy set, SAGE V2 with direction training achieves 58.3% — closing 62% of the gap with GloVe (83.3%). A sequence awareness experiment shows all four encoding modes achieve 100% recall on 12 agent navigation scenarios, and that sequence encoding is architecturally benign.

SAGE is not proposed as a replacement for parametric models in fitting-accuracy tasks. It is proposed as a replacement for parametric memory specifically — the component of AI systems responsible for storing and retrieving knowledge, which currently requires retraining to update and suffers catastrophic forgetting by design. The geometry computes; the weights are not needed.

Future work includes: (1) re-testing sequence encoding at higher memory density to measure encoding benefit; (2) the GeoMemFormer extension replacing transformer attention with geometric cube queries; (3) integration with vision and sensor encoders for real-world inputs beyond text embeddings; (4) direction training with category-specific relation pairs and early-stopping to improve full-benchmark analogy performance.

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